

using (*)

and taking $h = dx = (dx_1, \dots, dx_n)$,

we obtain $df(a)(dx) = \nabla f(a) \cdot dx$,

$$\text{or } df(a)(dx) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) x_i,$$

which is usually written in the form:

$$df(a) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) dx_i.$$

Using $d^k f(a)(h) \stackrel{\text{def}}{=} \frac{d}{dt} (d^{k-1} f(a+th)(h)) \Big|_{t=0}$,

we obtain:

$$d^k f(a) = \left(dx_1 \frac{\partial}{\partial x_1} + \dots + dx_n \frac{\partial}{\partial x_n} \right)^k f(a) \quad (\text{diferențială de ordin } k)$$

E20 If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ has continuous partial derivatives of the second order, then:

$$\begin{aligned} d^2 f &= \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} \right)^2 \\ &= \frac{\partial^2 f}{\partial x^2} dx^2 + \frac{\partial^2 f}{\partial y^2} dy^2 + 2 \frac{\partial^2 f}{\partial x \partial y} dx dy \quad (\text{formula clasa a 6-a}) \end{aligned}$$

Taylor's Formula for Functions of Several Variables

R23 Using ∇ operator, Taylor's formula can be written in the form.

$$f(x) = f(a) + \frac{((x-a)\nabla)f(a)}{1!} + \dots + \frac{((x-a)\nabla)^m f(a)}{m!} + \frac{((x-a)\nabla)^{m+1} f(\xi)}{(m+1)!}$$

Functional Sequences and Series

Functional Sequences

Let $A \subseteq \mathbb{R}$. Consider a functional sequence (f_n) of real functions defined on A and $f: A \rightarrow \mathbb{R}$

D1 A functional sequence (f_n) is convergent to f on A , and denoted by $f_n \rightarrow f$, if $\lim_{n \rightarrow \infty} f_n(x) = f(x)$, $(\forall) x \in A$. (pointwise convergence)

D2. A functional sequence (f_n) is said to be uniformly convergent to f on A , denoted by:

$$f_n \xrightarrow{u} f$$

if $(\forall) \varepsilon > 0$ $(\exists) \varepsilon$ s.t. $|f_n(x) - f(x)| < \varepsilon$ $(\forall) n \in \mathbb{N}$, $n > n_\varepsilon$; $(\forall) x \in A$.